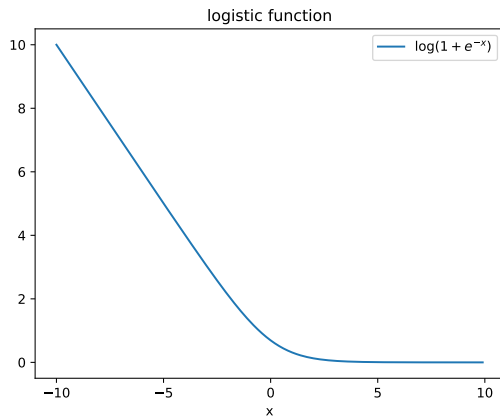


Fondamentaux théoriques du machine learning



First principal component

We look for w , $\|w\| = 1$ such that

$$\sum_{i=1}^n (w^T x_i)^2 \quad (1)$$

is maximal.

Proposition

w is the eigenvector of $X^T X$ with largest eigenvalue $\lambda_{\max}$.

First principal component

We look for w , $\|w\| = 1$ such that

$$\sum_{i=1}^n (w^T x_i)^2 \quad (2)$$

is maximal.

Proposition

w is the eigenvector of $X^T X$ with largest eigenvalue $\lambda_{\max}$.

Exercise 1: Show the proposition.

First principal component

$$\begin{aligned}\sum_{i=1}^n (w^T x_i)^2 &= \|Xw\|^2 \\ &= \langle Xw, Xw \rangle \\ &= \langle (X^T X)w, w \rangle\end{aligned}$$

This quantity is always smaller than $\lambda_{\max}$, and it is attained for an eigenvector in the eigenspace with norm 1, since we impose that $\|w\| = 1$.